

RAHUL ROY, MS

Software Development Engineer | Experienced in Cloud Technologies and Database Management, skilled at building and developing readable, maintainable, testable, scalable, reusable, and efficient code within Agile environments. Committed to innovation, collaboration, and continuous learning to drive technological advancements.

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in rahul-roy

Open to Relocation

TECHNICAL SKILLS

Languages: Java, Python, C/C++, SQL, Ruby, HTML, CSS, Kotlin, NoSQL, JavaScript, TypeScript, C#

Technologies: Spring Framework, REST APIs, Flutter, Angular, Node.js, Spark, Kafka, Hadoop, Hive, Firebase, AWS, Azure, Kubernetes, .net

Development Tools: Git, GitLab, Docker, Gradle, Maven, JWT, Heroku, Tomcat, Nginx, Jenkins, TravisCI, Junit5, WordPress, Jira, Trello

Databases: RDBMS (PostgreSQL, DB2, MySQL), NoSQL (MongoDB, Cassandra, DynamoDB)

Visualisation Tools: Tableau, Power BI, Microsoft BI, Adobe Creative Cloud, Final Cut

WORK EXPERIENCE

Software Engineer | Microsoft | Seattle, WA | July 2023 – Present

- Enhanced PostgreSQL database performance by 47% and cut query response times by 61% through strategic implementation of partial and multi-column indexing, optimizing database queries for enhanced efficiency.
- Implemented token-based credentialing microservice for *Georgia Gateway*, facilitating the onboarding of large-scale users (400,000+) by utilizing Spring Boot MVC and JSON Web Tokens (JWT) for backend services which improved 10% compatibility.
- Orchestrated front-end design and development of *PowerBI* dashboards, achieving a 35% improvement in issue resolution and business performance for clients by creating detailed design documents and integrating innovative visualization methods.

Research Assistant | [Dr. Edward Gehringer](#) | Carnegie Mellon University | Pittsburgh, PA | August 2022 – May 2023

- Directed the enhancement of the *Peer Assessment System - Expertiza*, by driving the transition to Role-Based Reviewing using Ruby on Rails and JavaScript, optimizing peer evaluation precision by 20% while ensuring system continuity.
- Piloted code refactoring initiatives utilizing Object Oriented principles and Docker to bolster *Expertiza*'s test coverage by 25% and code quality by 40%, fostering a more robust system embraced by 50+ open-source contributors.

Software Engineer Intern | Google | San Jose, CA | May 2022 – August 2022

- Improved *Zscaler Private Access (ZPA)* availability by 25% by instituting a throttling mechanism with Java and Spring, streamlining request management during failures and elevating the dependability of *Zero Trust Network Access (ZTNA)*.
- Restructured *ZPA* backend APIs by transitioning from asynchronous REST to GraphQL, enhancing system scalability and efficiency by 24%.
- Spearheaded automated and comprehensive E2E test scripts using TestNG, JUnit, and Mockito resulting in a 90% reduction in testing times.

Software Engineer | Amazon | Delhi NCR, India | July 2020 – August 2021

- Integrated and deployed Machine Learning models and rules in Java to detect potential passwords
- Achieved immense reduction of false positives across 100+ GitHub repositories and Teams
- Designed database schema and single-handedly coded application to use Cassandra end-to-end
- Upgraded 6 repositories to run on JDK 17, and wrote unit tests to increase code coverage by 50%

PROJECTS

Client Server File Transfer Application (Computer Networks)

- Architected a multi-threaded synchronized Python FTP Server and Client structure leveraging Socket Programming and ARQ schemes, resulting in a 30% increase in packet transmission reliability and a 25% reduction in data re-transmissions.
- Improvise packet transfer speed by 50% through the implementation of a multithreaded model that supports asynchronous execution, optimizing ARQ schemes, and fine-tuning key parameters, resulting in a 20% boost in transmission efficiency and 15% reduction in packet loss.

Lung Tuberculosis Detection (Deep Learning)

- Engineered deep learning models (UNet, LinkNet, FPNs) with anti-aliased CNNs in Python to improve tuberculosis detection accuracy by 10%, with findings authored in the Elsevier Procedia Computer Science journal.

Maze Solver (AI)

- Developed intelligent maze-solving agents for complex grid-based (51x51) environments with blocked cells and ghost obstacles, employing diverse navigation strategies to avoid ghosts
- Executed an optimal A* search with a heuristic derived from a relaxed version of the problem, and a Monte Carlo approach, accessing child node utilities by averaging the results of 100 simulations (1 for success, 0 for failure)
- Achieved survival rates of 76% for 5 Ghosts, 45% for 25 ghosts and 23% for 45 Ghosts

EDUCATION

North Carolina State University | Raleigh, NC, USA | August 2021 – May 2023

Master of Computer Science | GPA **4.00**

Coursework: Analysis of Algorithms, Internet Protocols, Object-Oriented Design and Development, Resource Dependent NN's, Automated Learning and Data Analysis

Guru Gobind Singh Indraprastha University | New Delhi, India | August 2016 – June 2020

B.Tech in Computer Science and Engineering | GPA **8.43**

Coursework: Computer Networks, Operating Systems, Data Structures, Software Engineering, Database Systems

LEADERSHIP & ACHIEVEMENTS

- Delivered a presentation on "Prediction of Cervical Cancer Using Chicken Swarm Optimization" at the International Conference on Innovations in Computer Science and Engineering (ICICC), hosted by VŠB – TUO, Czech Republic.
- Guided a transformative educational workshop teaching Android Application Development to over 50 high school students at Ryan International through the Atal Tinkering Lab (Govt. of India) program, fostering technological skills and innovation among youth.
- Attained 1st & 2nd grade in Plectrum Guitar from Trinity, London, UK, demonstrating dedication and skill in musical arts.
- Secured 1st place in Engineering Graphics Board Exam, meriting a certificate of Honour for outstanding performance.